Bioinformatics, 38(19), 2022, 4645–4646
https://doi.org/10.1093/bioinformatics/btac573
Advance Access Publication Date: 23 August 2022
Applications Note

OXFORD

Structural bioinformatics

YAMACS: a graphical interface for GROMACS

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Associate Editor: Alfonso Valencia

Received on May 10, 2022; editorial decision on August 19, 2022; accepted on August 22, 2022

Abstract

Summary: A graphical user interface for the GROMACS program has been developed as plugins for YASARA molecular graphics suite. The most significant GROMACS methods can be run entirely via a windowed menu system, and the results are shown on screen in real time.

Availability and implementation: YAMACS is written in Python and is freely available for download at https://github.com/YAMACS-SML/YAMACS and is supported on Linux. It has been released under GPL-3.0 license.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1. Introduction

Molecular dynamics (MD) is a powerful tool used in chemistry and biophysics to study the evolution of molecular systems, predicting their properties from the inherent interactions. GROMACS (Abraham *et al.*, 2015) is one of the golden standards in MD. It is particularly interesting to researchers since it was developed as open-source software and is freely available without restrictions.

GROMACS is run from the command line with user-provided configuration files. The lack of a graphical user interface (GUI) and the steep learning curve can distract the researcher from addressing the chemical or biological problem. To increase its usability, many core functionalities of GROMACS have become available through a complete GUI as a suite of plugins for YASARA molecular viewer (Krieger and Vriend, 2014), called YAMACS. YAMACS greatly simplifies the several preparation steps involved in setting up MD simulations in GROMACS. In addition, the suite allows the interactive display of MD simulations, a feature that is absent in GROMACS.

Several approaches to the development of a GUI for GROMACS have already been carried out in the past (Kagami *et al.*, 2017; Kota, 2007; Makarewicz and Kaźmierkiewicz, 2016; Sellis *et al.*, 2009), but the projects have, to the authors' knowledge, not been updated for several years at the time of submission and offer a limited number of options compared to YAMACS.

YAMACS plugins are available for YASARA View (free software) and YASARA Structure (proprietary software). YAMACS allows performing system preparation and running MD simulations using GROMACS engines through YASARA, with an easy-to-use windowed system. The complete trajectories and the results of GROMACS analysis are then displayed in YASARA. Currently, YAMACS is composed of five plugins, ranging from system preparation to selecting atoms to perform further calculations in GROMACS. The functionalities will be extended further, integrating new GROMACS functions into the suite. The scripts are available on GitHub, and users have the opportunity to introduce changes and new functions. Our goal is to build a strong community that can integrate new functions into YAMACS and pushes development to a higher level.

2. Implementation and required software

The plugins are written in Python programming language. The suite requires YASARA (View or Structure), GROMACS (2020.4 or higher) and Python (2.7 or higher). YASARA and GROMACS have to be installed separately. The plugins, being based on GROMACS, are available for Linux only. YASARA can be downloaded at http://www.yasara.org. GROMACS can be downloaded from https://www.gromacs.org. Python is installed by default on Linux machines.

YAMACS is available for download on https://github.com/YAMACS-SML/YAMACS and is released under GPL-3.0 license. The files must be extracted in the 'Plg' folder inside the YASARA installation path to install the plugin. Installing the plugins adds a 'YAMACS' dropdown menu in YASARA.

3. Documentation

Documentation is provided on YAMACS GitHub page. The repository offers regular plugin updates, step-by-step installation instructions, video tutorials and a detailed user manual. Tutorials based on recently published molecular dynamics studies are included (Lopez *et al.*, 2013; Piotto *et al.*, 2014, 2017; Scrima *et al.*, 2014; Sessa *et al.*, 2021). Instructions for implementing new plugins in YASARA

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can be found on the YAMACS GitHub page and YASARA user manual.

4. YAMACS plugins

Unlike other GUIs, limited to protein systems only, YAMACS can perform MD simulations for different systems: protein, protein–ligand complex, membrane, membrane–protein, membrane–protein–ligand and small molecules. The software is composed of five plugins:

1. **System preparation** (*YAMACS_1_PDB2GMX.py*): Using this plugin, the files needed to start an MD simulation will be generated. The user is given the option to choose the type of system and the force field.
2. **System equilibration** (*YAMACS_2_equilibration.py*): This plugin performs water and ion addition, energy minimization, NVT and NPT equilibration of the system.
3. **MD Run** (*YAMACS_3_MD.py*): This plugin is used to run MD simulations of the system, with user-defined simulation time. Several options are present; see user manual for more information.
4. **MD play** (*YAMACS_4_MdPlay.py*): This plugin allows to visualize GROMACS simulation trajectory in YASARA.
5. **Select Atom** (*YAMACS_5_Select_atom.py*): This plugin allows the selection of atoms, generating a *yas_index.ndx* file that GROMACS can read to perform further analyses. The implementation of new analyses will be made available on the GitHub page.

5. Summary

GROMACS is a highly comprehensive molecular simulation package. Its use to a broad audience is limited by the absence of an intuitive graphical interface. If the use of command line programs requires and stimulates the user's attention and full awareness of the tools, the presence of a graphical interface to interactively evaluate the progress of a calculation also allows to quickly identify anomalies or settings errors. The graphical interface for GROMACS eliminates manual writing of configuration files and the use of a command line interpreter. All required softwares are free of charge. It is expected that the plugins will attract non-experienced computer users to perform complex energy calculations with GROMACS at the touch of a button. YAMACS offers the possibility for all researchers to perform GROMACS simulations easily, simplifying the steps needed to perform and visualize the calculations. YAMACS has been designed with maximum ease of use in mind; however, the preparation of the simulation environment may still be complex, and users may encounter errors to resolve. The purpose of the GitHub community is to solve any errors that users may encounter quickly and offer everyone the possibility of modifying and extending the code by adding more functions to the suite, furthering the integration between YAMACS and GROMACS.

Acknowledgements

Special thanks to Florian Dommert and Stephen Lloyd Watkins, who helped us in the very early stages of the development.

Funding

This work was been supported by the University of Salerno.

Conflict of Interest: none declared.

References

Abraham,M.J. *et al.* (2015) GROMACS: high performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX*, **1-2**, 19–25.

Kagami,L.P. *et al.* (2017) LiGRO: a graphical user interface for protein-ligand molecular dynamics. *J. Mol. Model*, **23**, 304.

Kota,P. (2007) GUIMACS - a java based front end for GROMACS. *In Silico Biol.*, **7**, 95–99.

Krieger,E. and Vriend,G. (2014) YASARA view - molecular graphics for all devices - from smartphones to workstations. *Bioinformatics*, **30**, 2981–2982.

Lopez,D.H. *et al.* (2013) 2-Hydroxy arachidonic acid: a new non-steroidal anti-inflammatory drug. *PLoS One*, **8**, e72052.

Makarewicz,T. and Kaźmierkiewicz,R. (2016) Improvements in GROMACS plugin for PyMOL including implicit solvent simulations and displaying results of PCA analysis. *J. Mol. Model.*, **22**, 109.

Piotto,S. *et al.* (2014) Differential effect of 2-hydroxyoleic acid enantiomers on protein (sphingomyelin synthase) and lipid (membrane) targets. *Biochim. Biophys. Acta*, **1838**, 1628–1637.

Piotto,S. *et al.* (2017) Computational study on human sphingomyelin synthase 1 (hSMS1). *Biochim. Biophys. Acta. Biomembr.*, **1859**, 1517–1525.

Scrima,M. *et al.* (2014) Structural features of the C8 antiviral peptide in a membrane-mimicking environment. *Biochim. Biophys. Acta*, **1838**, 1010–1018.

Sellis,D. *et al.* (2009) Gromita: a fully integrated graphical user interface to gromacs 4. *Bioinform. Biol. Insights*, **3**, 99–102.

Sessa,L. *et al.* (2021) Hydroxylated fatty acids: the role of the sphingomyelin synthase and the origin of selectivity. *Membranes*, **11**, 787.

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